Yizhe Dai

Data Engineering — data quality & evaluation frameworks · schema design & multi-source integration · governance & lineage

Seeking a **Summer 2027 Data Engineering internship** · available May–Aug. 2027

Email: yizhed7@gmail.com | LinkedIn: linkedin.com/in/yizhe-dai-61b83a369 | Boston, MA, USA

Education

Harvard University — Aug. 2026 – Expected May 2028
M.S. Health Data Science, T.H. Chan School of Public Health — *Boston, Massachusetts, USA*

University of Michigan, Ann Arbor — Aug. 2024 – May 2026
B.S.E. Industrial and Operations Engineering; Minor in Computer Science — *Ann Arbor, Michigan, USA*

UM–SJTU Joint Institute, Shanghai Jiao Tong University — Sep. 2022 – Aug. 2026
B.S.E. Electrical and Computer Engineering — *Shanghai, China*

Dual degree: Michigan and SJTU are one joint program; dates overlap by design

Coursework: Database Systems, Data Structures & Algorithms, Engineering Data Analytics, Probability & Statistics

Skills

Languages: Python, SQL, TypeScript

Data & databases: PostgreSQL · dbt (metadata & lineage tooling) · gold-case SQL regression and contract tests · time-series feature engineering, missing-value imputation, cleaning rules for high-volatility series

Frameworks & Tools: FastAPI, Flask, React, Playwright/CDP, Git

Experience

Lark, ByteDance (enterprise collaboration & AI suite) — May 2026 – Aug. 2026
AI Solutions Intern, Tech Stack: Python, Flask, TypeScript, Playwright/CDP — *Shanghai, China*

- Designed a unified **trace schema** that made two enterprise AI agents comparable on real user tasks: drove both Electron clients over Chrome DevTools Protocol to extract their telemetry, mapped two incompatible event vocabularies onto one field set, and scored the normalized runs side by side.

Shenzhen Legion Technology Co., Ltd. — Apr. 2025 – Oct. 2025
Data Engineering & AI Intern (part-time), Tech Stack: Python, SQL, dbt, RAG — *Remote, China*

- Built the **evaluation framework** for a RAG data-governance copilot serving dbt pipelines and lineage: structured cases across its three question classes (catalog, SQL, data-QA), each graded on whether retrieval returned the right catalog/lineage objects and whether the answer met governance rules, so prompt and retrieval changes were re-scored on the same cases, not eyeballed.
- Designed the copilot's prompt architecture: the classifier behind those classes, entity extraction, and embedding-based search.

University of Michigan — Cardiac Rehabilitation Wearable Study — Apr. 2025 – Jan. 2026
Research Assistant (part-time) — SURE summer program, then continued on the study — *Ann Arbor, Michigan, USA*

- Assembled the cardiac-rehabilitation analysis dataset: reconciled 400K+ raw step records from multiple sources onto one per-patient daily timeline and set missing-value and cleaning rules that cut 247 enrolled patients to a **191-patient analysis cohort**.
- Engineered temporal features (rolling averages, trend slope, periodicity) feeding K-means/PCA phenotyping and Kalman-filter plateau detection against the standard 36-session model; **∼68%** of that 191-patient cohort had suboptimal discharge timing.

NeuRRo Lab, University of Michigan — Oct. 2024 – Apr. 2026
Research Assistant (part-time), Tech Stack: MATLAB, Python/PyTorch, Unity, C# — *Ann Arbor, Michigan, USA*

- Built the lab's real-time EMG biofeedback system and MATLAB muscle-synergy analysis code, both used in clinical experiments.

Projects

MetaFlow (two-person project) – *metadata-centric NL-to-SQL; Python, SQL, PostgreSQL* — Mar. 2026 – Present

- Own the metadata and evaluation layer: **20 gold-case scenarios** for SQL regression, each pinned across all five pipeline stages as version-controlled expectations, so a metadata or schema change is caught by re-running the suite, not spot-checked by hand.
- Build the FastAPI backend on **PostgreSQL** — including the contracts endpoint that enforces data-quality rules as contract-as-test — and the React/TypeScript studio front end for a system that resolves natural language into SQL through an inspectable relational IR rather than emitting SQL directly.

LLM Data Flywheel for In-Vehicle Voice Assistants – *Undergraduate Thesis; Python* — Jun.–Aug. 2026

- Owned the data pillar of a multi-model in-vehicle voice assistant: a user simulator (41 persona archetypes, explicit state control) with retrieval-grounded rollouts, LLM-judge scoring, and a deterministic governance layer yielded **17,200** verified (reward=1) fine-tuning dialogues, with targeted error-regeneration on the 11 highest-error functions.
- Fine-tuning on this corpus raised judge-adjusted answer acceptance **72.8%→82.7%**, lifted the strict multi-turn all-correct rate 39.9%→60.0%, and cut no-tool false positives 4,424→1,410.